

AI-POWERED · AGENTIC · OPEN SOURCE

Supply Chain Health Agent

A step-by-step guide to diagnosing, improving,
and future-proofing your supply chain with AI

For: Supply Chain Leaders, C-Suite & Board Members

May 2026 | supply-chain-health-agent.streamlit.app

60s

Time to
assessment

8

Supply chain
domains

100%

Browser-based,
no setup

SECTION 01

The Challenge

Supply chain failures are costing enterprises more than ever — yet diagnosis still takes months and costs hundreds of thousands of dollars

Setting the context for why a faster, smarter tool is needed

Supply chain underperformance costs the average enterprise \$100M–\$300M annually — most of it invisible until it is too late

\$182M

avg. annual loss

per Fortune 500 from supply chain disruptions (McKinsey, 2024)

12 wks

avg. assessment time

for a traditional consulting-led supply chain diagnostic

\$300K

avg. consulting cost

for a full-scope supply chain health assessment

94%

of executives

say supply chain visibility is their #1 concern (Gartner, 2025)

The gap between world-class and average supply chain performance

Domain	World-class	Average	Annual gap
OTIF	95–98%	85–88%	~\$12M/yr penalties & premium freight
Inventory Turns	8–12x	4–5x	~\$40M trapped per \$200M inventory
Forecast Accuracy	85–90%	60–70%	25–35% excess safety stock cost
OEE	>85%	60–65%	~\$18M lost output per \$100M capacity

SECTION 02

The Tool

What the Supply Chain Health Agent is, how it works, and who it is designed for

A plain-language introduction before the step-by-step walkthrough

The Supply Chain Health Agent delivers in 60 seconds what traditionally takes 12 weeks — browser-based, no technical setup required

What it is

An AI-powered supply chain diagnostic that scores your organisation across 8 domains, benchmarks against 2025 world-class standards, and delivers a prioritised action plan — in under 60 seconds.

Instant Diagnostics

Scores 8 domains 0–100 vs 2025 Gartner & SCOR benchmarks

Live Market Intelligence

Searches the web in real time for freight rates, regulatory changes

Auto Action Pack

Generates board summary, 90-day plan & risk list without being asked

Cost-Benefit Analysis

Sizes the dollar cost of every gap using your real financial data

8

domains scored

8

industry verticals

2

user tracks

60s

to first insight

100%

browser-based

Free

open source MIT

All outputs are illustrative and directional. Not professional financial advice. Financial data entered by users is never stored.

Three steps from opening a browser to having a board-ready assessment: Select, Assess, and Act

1

SELECT

Choose your profile

The tool asks who you are — Supply Chain Manager or C-Suite Leader. Everything adapts: questions, language, depth, and output format.

2

ASSESS

Run your assessment

Answer a few questions or run an instant general benchmark. Claude searches the web for live market data then scores your supply chain vs 2025 world-class standards.

3

ACT

Receive your outputs

Results arrive in four tabs. Board summary, 90-day action plan, risk watch list, and strategic scenarios are generated automatically.

Privacy first: any financial data you enter is never stored. It exists only in your browser session and is discarded when you close the tab. GDPR & CCPA compliant.

Two completely different experiences — one tool. The experience forks from the very first screen based on who you are

ANALYST / MANAGER TRACK

- ✓ Deep KPI diagnostics across all 8 SCOR domains
- ✓ Custom domain inputs — describe your situation
- ✓ Benchmark gaps vs 2025 Gartner world-class thresholds
- ✓ Cost-benefit analysis with your real financial figures
- ✓ Follow-up Q&A with live web search
- ✓ For: Supply chain managers, analysts, operations teams

VS

C-SUITE / BOARD TRACK

- ✓ Strategic scenario comparison — 3 named paths with trade-offs
- ✓ 36-month phased maturity roadmap: Stabilise → Build → Lead
- ✓ Board-ready language — no jargon, impact-first framing
- ✓ Customisable orientation: growth vs cost-out vs balanced
- ✓ Financial context for dollar-sized scenario estimates
- ✓ For: CEOs, CFOs, COOs, board members

Eight domains give a complete picture of supply chain health — from customer demand to ESG, nothing is left unassessed

Demand Planning

Forecast accuracy, bias, MAPE, S&OP maturity. World-class: FA >85%

Procurement

Supplier OTD, spend under mgmt, PPM defects. World-class: OTD >95%

Manufacturing

OEE, first-pass yield, schedule adherence. World-class: OEE >85%

Inventory

Turns, stockout rate, E&O%, cash-to-cash. World-class: 8-12x turns

Logistics

OTIF, freight cost %, perfect order rate. World-class: OTIF 95-98%

Warehousing

Order accuracy, utilisation, lines/hour. World-class: >99.9% accuracy

Risk & Resilience

BCP coverage, TTR, supplier risk tiers. World-class: BCP >95% nodes

Sustainability & ESG

Scope 3 tracking, ESG audit coverage. World-class: >80% Tier-1 audited

Score bands:



80-100 Excellent — World-class



60-79 Good — Above avg



40-59 Fair — Needs work



0-39 At Risk — Act now

SECTION 03

Step-by-Step Walkthrough

A plain-language guide to every screen and every output — from first click to board-ready results

Six steps, zero technical knowledge required

Step 1 — Select your profile: the very first screen asks who you are, and adapts everything from that single choice

1

Landing screen — persona selection



Supply Chain Manager / Analyst

Deep KPI diagnostics, benchmarks, cost-benefit analysis. Built for practitioners.

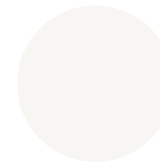
PRACTITIONER TRACK



C-Suite / Board Member

Strategic scenarios, maturity roadmap, board-ready outputs.

EXECUTIVE TRACK



Enterprise Configuration

Embed within your organisation's systems with custom controls.

COMING SOON

Step 2 — Configure: choose your industry vertical and whether to run an instant benchmark or a custom deep-dive

2

Sidebar configuration — takes 30 seconds

Choose your industry

Each vertical applies sector-specific benchmarks, risk weighting, and reference organisations

General

Semiconductor

Automotive

Pharmaceutical

Retail & E-commerce

Consumer Packaged Goods

Aerospace & Defense

Healthcare

Choose your assessment mode

General Benchmark

(Instant — no inputs needed)

Claude benchmarks against typical industry performance for your vertical. Ideal for a fast peer comparison — 60 seconds, zero preparation.

Custom Assessment

(Tailored — describe your org)

You describe your organisation across each domain. Claude gives a far more accurate personalised diagnosis. 5–10 minutes of input, meaningfully richer output.

Step 3 — Describe your organisation: 8 plain-language prompts, or skip entirely for an instant benchmark

3

What to tell Claude — no jargon required

Demand Planning

"We have 68% forecast accuracy and run S&OP monthly. No AI yet."

Manufacturing

"Our OEE is approximately 65%. We run two shifts. Some changeover issues."

Logistics

"OTIF is 88%. We use 3PLs for all outbound. Freight is 7% of revenue."

Risk & Resilience

"We have BCP for top 10 suppliers. No formal resilience score yet."

Procurement

"We have 180 active suppliers. OTD is around 82%. Some single-source risk."

Inventory

"We carry about 75 days of inventory. Turns are 5x. Some excess stock."

Warehousing

"Order accuracy is 97%. We pick manually. Two DCs, 80% utilised."

Sustainability

"We track Scope 1 and 2. Scope 3 limited. ESG audit coverage is 40%."

Skip tip: leave any field blank and Claude will note the gap and use industry benchmarks instead. Run a full assessment with zero inputs.

Step 4 — Your results arrive in four tabs: Diagnostic Report, Action Pack, Market Intelligence, and Q&A

4

Results page — everything in one place

69

Overall score

GOOD

Semiconductor vertical

Demand	72/100
Procurement	65/100
Manufacturing	78/100
Inventory	68/100
Logistics	71/100
Warehousing	74/100
Risk	58/100
Sustainability	63/100



Diagnostic Report

Diagnostic Report

Executive summary, top risks, domain highlights, priority recommendations, cost-benefit analysis.

Action Pack

Action Pack

Board summary (5 bullets), 90-day action plan with owners, top-5 risk watch list.

Market Intelligence

Market Intelligence

5 real-time supply chain signals searched during your assessment — freight, regulations, risks.

Q&A

Q&A

Ask Claude anything. Suggested questions provided. Claude can search the web for current data.

Step 5 — Your action pack is generated automatically: board summary, 90-day plan, and risk watch list — no prompting required

5

Agentic AI — works without being asked

Board Summary

5 bullets. Written for CFO / CEO. Impact-first.

Overall score 69/100 — Good, 2 domains At Risk

Risk & Resilience gap costs est. \$8M–\$15M/yr

Manufacturing gap (OEE 65% vs 85%) = est. \$12M lost

Priority: close Risk and Logistics gaps — payback <12 months

Recommended: Selective Transformation — \$3M–\$6M, 18 months

90-Day Action Plan

8–10 specific actions with owners, metrics, timing.

Wk 1–4: Map all Tier-1 & Tier-2 suppliers — Procurement

Wk 1–4: Implement export control monitoring — Compliance

Wk 5–8: Qualify 2 alternative suppliers for single-source parts

Wk 5–8: Launch AI demand sensing pilot — Supply Planning

Wk 9–12: Deploy real-time OTIF dashboard — Logistics Mgr

Risk Watch List

Top 5 risks: severity, likelihood, immediate action.

HIGH Supply concentration >60% Taiwan-based — diversify

HIGH Export compliance 72hr vs <24hr target — automate

MED Forecast accuracy 68% — below 75% benchmark

MED Scope 3 <30% tracked — EU CSDDD risk by 2026

LOW Warehouse utilisation 88% — congestion threshold

These three outputs are generated automatically — you do not need to ask for them. This is what makes the tool truly agentic.

Step 6 — Ask Claude anything: live Q&A with web search access, plus strategic scenarios and roadmap for C-Suite users

6 Interactive follow-up — the conversation continues

Follow-up Q&A — all users

"What are the top 3 actions for next 90 days?"

→ Specific actions with timelines and named owners

"How does our OTIF compare to Walmart's mandate?"

→ Claude searches live data and benchmarks your rate

"What's the cost of staying at our current OEE?"

→ Calculates annual revenue impact of the gap

"Board-level summary in 3 bullet points please"

→ Concise, jargon-free executive summary instantly

Executive track extras — C-Suite only

3 Strategic Scenarios

Named paths (e.g. 'Optimise Now' vs 'Selective Transformation' vs 'Full Digital') each with investment range, expected KPI movement, key trade-offs, and a recommended choice.

36-Month Maturity Roadmap

Four phases: Now → Stabilise (0–6 months) → Build (6–18 months) → Lead (18–36 months). Each with specific initiatives, investment range, and measurable milestones.

Customisation Options

Adjust assessment orientation: Growth-first vs Cost-out vs Balanced. Set your timeline preference and benchmark peer before running.

SECTION 04

Expected Outcomes

What users walk away with — business value, financial impact, and strategic advantage

From diagnostic to decision-ready in a single session

Users walk away with six decision-enabling outputs — everything needed to brief leadership, prioritise investment, and act

A scored, benchmarked health check

Domain scores 0–100 against 2025 world-class. You know exactly where you stand vs Gartner Top 25.

The dollar cost of every gap

Each domain gap is sized in financial terms — working capital trapped, penalty exposure, lost revenue.

A board-ready executive summary

5 bullets for CFO or CEO. Impact-first, no jargon. Ready to paste into a board presentation.

A prioritised 90-day action plan

Named initiatives, suggested owners, success metrics, and week-by-week timing. Act on it immediately.

A risk watch list

Top 5 supply chain risks by severity and likelihood, each with an immediate mitigation action.

Strategic scenarios (C-Suite track)

Three named transformation paths with investment ranges, KPI targets, trade-offs, and a recommendation.

The cost-benefit analysis turns a diagnostic into a business case — sized to your real financial data, not industry generics

How the CBA works

- 1 Enter your data**
Revenue, inventory value, OTIF rate, freight %, gross margin. Optional — blanks use benchmark assumptions.
- 2 Claude sizes each gap**
For every domain below 70, Claude calculates the annual cost of the performance gap using your figures.
- 3 Benefit of closing gap**
Claude estimates the annual recurring upside once performance reaches world-class — as a dollar range.
- 4 Implementation estimate**
Low / Medium / High effort with indicative cost range (\$, \$\$, \$\$\$) and expected timeline to full benefit.
- 5 Priority matrix**
Domains ranked by ROI: Quick Wins (<12 month payback), Strategic Bets, Maintenance, or Deprioritise.

Illustrative CBA output — Risk & Resilience domain

Item	Value	Basis
Cost of current gap	\$8M– \$15M/year	Disruption exposure at current Risk score of 58
Benefit of closing gap	\$6M– \$12M/year	Recurring annual saving once Risk reaches 80+
Implementation investment	\$500K–\$2M (\$\$)	Supplier mapping, BCP update, tools, 12–18 months
Estimated payback	12–18 months	Midpoint cost divided by midpoint annual benefit
Priority tier	Quick Win	High benefit, medium effort, payback under 18 months

All CBA figures are directional estimates for illustrative purposes only. Not certified financial projections. Validate with your finance team before investment decisions.

The Supply Chain Health Agent outperforms traditional diagnostics on every dimension that matters to a leader

Dimension	Traditional Consulting	Internal Team Assessment	Supply Chain Health Agent
Time to first insight	8–12 weeks	4–8 weeks	Under 60 seconds
Cost	\$150K–\$400K	\$30K–\$80K time	Free (open source)
Live market data	Static at engagement	Manual research	Live web search every run
KPI benchmarks	Varies by consultant	Internal benchmarks	2025 Gartner + SCOR
Action plan	In final report	Manually compiled	Auto-generated, instant
Board-ready summary	Separate item	Manual write-up	Auto-generated, instant
Financial sizing	Separate workstream	If finance engaged	Integrated, uses your data
Available 24/7	Fixed engagement	Staff dependent	Always on, always current
Privacy	Data shared widely	Internal only	Zero storage, GDPR compliant

SECTION 05

By Industry

How the tool adapts to your sector's specific benchmarks, risks, and operational realities

Four industry groupings covering all eight supported verticals

Manufacturing & Automotive: OEE, zero line stoppages, and EV transition readiness — benchmarked against Toyota TPS and BMW

General Manufacturing

Reference: Gartner Top 25

OEE	>85%	60-65%	Lean/Six Sigma drives world-class OEE
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Schedule adherence	>95%	80-88%	Digital twins simulate before execution
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Inventory turns	8-12x	4-6x	AI reducing safety stock 15-30%
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First-pass yield	>98%	92-95%	Real-time quality monitoring standard
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Scope 3 tracking	>80%	<40%	EU CSDDD compliance mandatory by 2026
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Automotive

Reference: Toyota TPS, BMW, Tesla

PPM defects	<50 PPM	200-500	IATF 16949 compliance non-negotiable
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Line stoppages	0	2-5/month	JIT/JIS delivery demands zero tolerance
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Supplier OTD	>98%	88-92%	EV supply chains: new battery supplier risk
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EV transition KPI	Defined	Undefined	Battery raw material risk is #1 new challenge
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BCP coverage	>95%	60-75%	Semiconductor and rare earth disruption focus
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Pharmaceutical & Healthcare: cold chain integrity, serialisation, and regulatory compliance — where supply chain failure costs lives

Pharmaceutical

Reference: Pfizer, Roche, Johnson & Johnson

Cold chain compliance

100%

95-98%

GDP compliance mandatory
— any break is product loss

Batch traceability

100%

90-95%

DSCSA serialisation now
required in US

Recall readiness

<2 hrs

12-48 hrs

FDA expects documented
recall capability

OEE (pharma)

>85%

40-60%

Regulated environment
suppresses OEE vs general
mfg

Forecast accuracy

>85%

60-70%

Demand sensing critical for
short-lifecycle products

Healthcare

Reference: Medtronic, Stryker, Becton Dickinson

Stockout rate (critical)

<0.5%

2-5%

UDI traceability mandatory
under FDA 21 CFR 820

Recall response time

<24 hrs

3-7 days

GPO compliance shapes
distributor relationships

Order accuracy

>99.5%

97-98%

Sterile supply chain requires
zero contamination risk

Supplier ESG audit

>80%

35-50%

Patient safety demands
end-to-end supplier
oversight

Inventory DIO

60-90 days

90-120
days

Regulatory hold
requirements inflate DIO vs
other sectors

Retail & CPG: OTIF, case fill, and omnichannel fulfillment — where Walmart's 3% COGS penalty for late delivery sets the bar

Retail & E-commerce

Reference: Amazon, Walmart, Zara

OTIF rate	98%	85-92%	Walmart mandates 98% — 3% COGS penalty for failure
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Inventory turns	8-15x	4-6x	Zara achieves >12x through fast fashion model
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Return rate	<15%	20-30%	E-commerce returns are #1 profitability challenge
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Forecast accuracy	>85%	60-70%	AI demand sensing reduces MAPE by 30-50%
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Order cycle time	Same day	3-5 days	Amazon effect has reset customer expectations globally
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Consumer Packaged Goods

Reference: P&G, Unilever, Nestle

Case fill rate	>98%	92-95%	Retailer deductions for shorts avg 1-3% of revenue
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On-shelf availability	>97%	92-94%	Out-of-stock costs FMCG brands ~4% of annual revenue
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Promo forecast bias	+2%	+15-25%	Trade promo accuracy is #1 forecast failure driver
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Packaging ESG score	Certified	Partial	Retailer sustainability mandates affect shelf placement
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Supplier compliance	>90%	70-80%	Retailer chargebacks for non-compliance avg 0.5-1% COGS
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Semiconductor & Aerospace: fab yield, export controls, and long-lead parts — the most geopolitically exposed supply chains of 2025

Semiconductor

Reference: TSMC, Intel, NVIDIA, ASML

Die yield	>92%	70-85%	Fab utilisation 85-95% — below is waste, above is risk
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Fab cycle time	<60 days	90-120 days	Leading-edge nodes extend cycle times significantly
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Critical material cover	180+ days	60-90 days	Rare earth and specialty gas concentration risk is #1
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Export compliance	<24 hrs	48-96 hrs	CHIPS Act and BIS EAR require near-real-time monitoring
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Supplier concentration	<20%	40-70%	Taiwan concentration risk is geopolitical priority 2025
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Aerospace & Defense

Reference: Boeing, Airbus, Lockheed Martin

On-time delivery	>95%	82-88%	AS9100D compliance — programme delivery is at stake
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Parts traceability	100%	90-95%	Serialisation required for every component, no exceptions
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Long-lead coverage	24+ months	12-18 months	Some parts have 12-36 month lead times — forecast must be perfect
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ITAR compliance	100%	100%	Export control failure = criminal liability, zero tolerance
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Defect rate	Zero defect	Near-zero	Safety-critical applications — no defect rate is acceptable
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SECTION 06

What's Next & Getting Started

Agentic AI capabilities, the Phase 2 roadmap, and three steps to your first assessment

The future is designed and visible in the app today

Agentic AI means Claude works autonomously — searching, synthesising, and acting without waiting to be asked

What makes it agentic — Phase 1 (live now)

Live Web Search

Claude autonomously searches for live market data, freight rates, regulatory updates, and geopolitical risks during every assessment — from right now, not training data.

Auto Action Pack

The board summary, 90-day action plan, and risk watch list are generated automatically the moment your assessment completes. No prompt needed.

Market Intelligence

A dedicated parallel search surfaces the 5 most important supply chain developments affecting your vertical today.

Agentic Q&A

Follow-up questions can trigger web searches — so 'what are spot freight rates right now?' gets a current answer rather than a guess.

Conversational AI vs Agentic AI

Conversational AI

Waits for every prompt

Responds from training data

One step at a time

Agentic AI

Acts autonomously

Searches the web now

Multiple steps without asking

Phase 2 — Designed and Coming

Cross-session memory — Claude tracks your KPI trends over time

Proactive alerts — flags risks before you ask

Workflow integrations — push to Slack, Jira, email from results

Enterprise configurator — embed with your branding and access controls

Getting started takes three steps and less than five minutes — no installation, no technical knowledge, no setup required

1

Open the app

supply-chain-health-agent.streamlit.app

Open any browser on desktop or laptop and go to the URL above. No login, no download, no IT ticket. The app loads in seconds.

2

Select your profile

30 seconds

Choose Supply Chain Manager / Analyst or C-Suite / Board Member. Select your industry vertical. Optionally adjust parameters.

3

Run your assessment

Under 60 seconds

Click Run Assessment. Claude searches the web, analyses your supply chain, and delivers the full diagnostic, action pack, and market intelligence.

What you will have:

✓ Domain scores 0–100

✓ Priority recommendations

✓ Board summary

✓ 90-day action plan

✓ Risk watch list

✓ Market intelligence

The supply chain leaders who win in 2025 are the ones who know their numbers.

Now you can know yours — in 60 seconds.



Try it now:

supply-chain-health-agent.streamlit.app



GitHub:

github.com/dwnjuguna/supply-chain-health-agent



Author: David Njuguna | MIT License — free to use, share, and adapt



Open the app →

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